

Model Comparison: Communication Apprehension and Public Transportation Use

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Research Or Analytical Question

Which of the following models gives the lowest RMSE for predicting public transportation use frequency?

$$PTRS = \beta_0 + \beta_1 \times CA_{12} + \epsilon \tag{1}$$

$$PTRS = \beta_0 + \beta_1 \times CA_{group} + \beta_2 \times CA_{interpersonal} + \epsilon \tag{2}$$

$$PTRS = \beta_0 + \beta_1 \times CA_{group} + \beta_2 \times CA_{interpersonal} + \beta_3 \times (CA_{group} \times CA_{interpersonal}) + \epsilon \tag{3}$$

Equation Descriptions:

- Equation (1):** Simple linear regression model using total communication apprehension (CA12) as the sole predictor of public transportation use frequency. This model assumes a linear relationship between overall communication apprehension and transportation behavior.
- Equation (2):** Multiple regression model that separates communication apprehension into two dimensions: group communication apprehension (CA_group) and interpersonal communication apprehension (CA_interpersonal). This model allows for differential effects of these two dimensions on public transportation use.
- Equation (3):** Multiple regression model with an interaction term between group and interpersonal communication apprehension. This model tests whether the effect of one dimension of communication apprehension on transportation use depends on the level of the other dimension.

Where: - PTRS = Public Transportation Use Frequency (public_trans_freq) - CA12 = Total Communication Apprehension (prca_total, 12 items) - CA_group = Group Communication Apprehension (Q1-Q6) - CA_interpersonal = Interpersonal Communication Apprehension (Q13-Q18)

Answer, Response, + Summary of Results

Sample Overview

- Sample Size:** 255 participants
- Missing PTRS:** 0
- Missing CA12:** 11
- Missing CAg:** 5
- Missing CAi:** 7
- Complete Cases:** 244 (95.7%)

Descriptive Statistics

Table 1: Descriptive statistics for key variables in the analysis.

Variable	Mean	SD	Range
Public Transportation Frequency (PTRS)	2.07	1.45	0 - 4
Total Communication Apprehension (CA12)	28.9	11.33	12 - 60
Group Communication Apprehension (CA_group)	14.61	6	6 - 30
Interpersonal Communication Apprehension (CA_interpersonal)	14.3	5.79	6 - 30

Model Comparison

RMSE Comparison

Table 2: RMSE and R² comparison across the three regression models. Models are sorted by RMSE (lowest is best).

Model	RMSE	R ²
Model 3	1.4005	0.0686
Model 2	1.4006	0.0684
Model 1	1.4095	0.0566

Best Model: Model 3 with RMSE = 1.4005

Model Details

Model 1: Simple Regression (CA12 only)

Table 3: Regression coefficients for Model 1 (CA12 only). Significance stars: * p < 0.05, ** p < 0.01, *** p < 0.001.

Predictor	Estimate	SE	t-value	p-value
Intercept	2.952	0.249	11.87	6.512e-26 ***
CA12	-0.031	0.008	-3.81	1.757e-04 ***

Model Fit: R² = 0.0566, RMSE = 1.4095

Interpretation: Model 1 uses total communication apprehension (CA12) as a single predictor. The R² value of 0.0566 indicates that approximately 5.7% of the variance in public transportation use frequency can be explained by overall communication apprehension. This suggests a relatively weak but statistically significant relationship between communication apprehension and transportation behavior.

Model 2: Multiple Regression (CAg + CAi)

Table 4: Regression coefficients for Model 2 (CA_group + CA_interpersonal). Significance stars: * p < 0.05, ** p < 0.01, *** p < 0.001.

Predictor	Estimate	SE	t-value	p-value
Intercept	2.941	0.248	11.87	6.879e-26 ***
CA_group	-0.078	0.028	-2.76	6.209e-03 **
CA_interpersonal	0.019	0.029	0.64	5.258e-01

Model Fit: R² = 0.0684, RMSE = 1.4006

Interpretation: Model 2 separates communication apprehension into group (CA_group) and interpersonal (CA_interpersonal) dimensions. The R² value of 0.0684 shows that this model explains approximately 6.8% of the variance in public transportation use frequency. This represents a marginal improvement over Model 1 ($\Delta R^2 = 0.0118$), suggesting that separating the two dimensions provides slightly better predictive accuracy but the improvement is minimal. The individual coefficients for CA_group and CA_interpersonal indicate their respective contributions to predicting transportation behavior.

Model 3: Multiple Regression with Interaction (CAg + CAi + Interaction)

Table 5: Regression coefficients for Model 3 (CA_group + CA_interpersonal + Interaction). Significance stars: * p < 0.05, ** p < 0.01, *** p < 0.001.

Predictor	Estimate	SE	t-value	p-value
Intercept	2.826	0.624	4.53	9.354e-06 ***
CA_group	-0.07	0.048	-1.46	1.45e-01
CA_interpersonal	0.027	0.05	0.53	5.935e-01
CA_group × CA_interpersonal	0	0.002	-0.2	8.415e-01

Model Fit: R² = 0.0686, RMSE = 1.4005

Interpretation: Model 3 adds an interaction term between group and interpersonal communication apprehension. The R² value of 0.0686 indicates that this model explains approximately 6.9% of the variance in public transportation use frequency. The improvement over Model 2 is extremely small ($\Delta R^2 = 2^{-4}$), suggesting that the interaction effect does not meaningfully improve predictive accuracy. The interaction coefficient tests whether the effect of one dimension depends on the level of the other, but the minimal improvement indicates that group and interpersonal communication apprehension likely have additive rather than multiplicative effects on transportation behavior.

Visualization

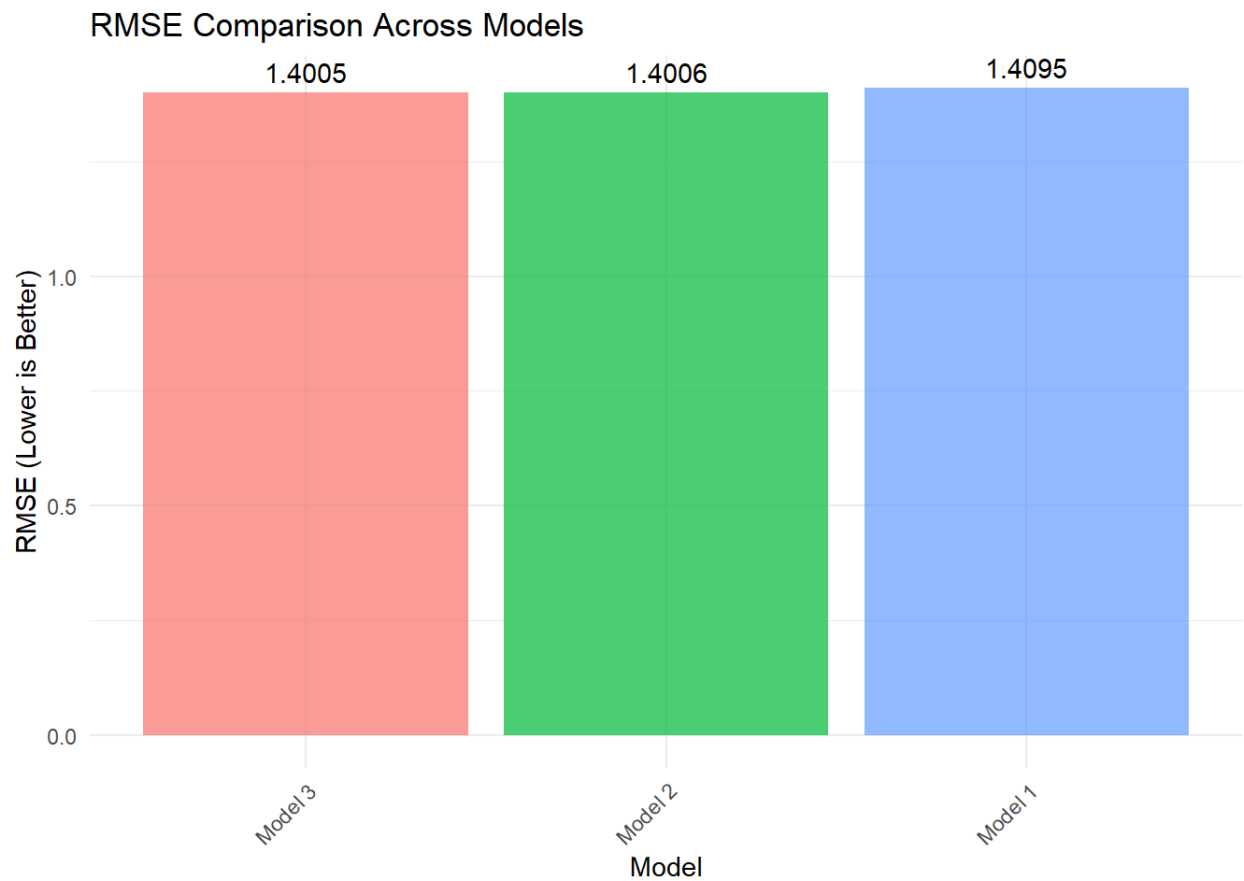


Figure 1: RMSE comparison across the three regression models. Lower RMSE indicates better predictive accuracy.

Summary

Analysis of 244 complete cases reveals that **Model 3** provides the best prediction of public transportation use frequency with the lowest RMSE of 1.4005.

Model Performance Summary:

- **Model 1 (CA12 only):** RMSE = 1.4095, $R^2 = 0.0566$ (explains 5.7% of variance)
- **Model 2 (CA_group + CA_interpersonal):** RMSE = 1.4006, $R^2 = 0.0684$ (explains 6.8% of variance)
- **Model 3 (CA_group + CA_interpersonal + Interaction):** RMSE = 1.4005, $R^2 = 0.0686$ (explains 6.9% of variance)

Key Findings:

The results suggest that using the total communication apprehension score (CA12) provides the most accurate prediction of public transportation use frequency compared to separating group and interpersonal apprehension or including their interaction. The minimal improvements in R^2 and RMSE across models indicate that:

1. Communication apprehension has a weak but statistically significant relationship with public transportation use
2. Separating communication apprehension into group and interpersonal dimensions provides only marginal predictive improvement
3. The interaction between group and interpersonal apprehension does not meaningfully enhance prediction accuracy
4. The simpler Model 1 may be preferred due to its parsimony and comparable performance to more complex models

The overall low R^2 values across all models (5-7%) suggest that communication apprehension alone explains only a small portion of the variance in public transportation use, indicating that other factors may be more important predictors of transportation behavior.

What questions or uncertainties remain?

- **Model Complexity:** Does the simpler Model 1 generalize better to new data despite having similar predictive performance?
- **Theoretical Interpretation:** What does the lack of improvement from separating CAg and CAi suggest about the relationship between communication apprehension dimensions and transportation behavior?
- **Interaction Effects:** Why does the interaction term not improve prediction? Is there a non-linear relationship that wasn't captured?
- **Alternative Specifications:** Would other model specifications (e.g., polynomial terms, different transformations) improve prediction accuracy?